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MILITARY SPECIFICATION

ROLLER, TOWED, PNEUMATIC TIRED, HOPPER BODY, OSCILLATING WHEELS

This specification is approved for use within CODE 84 Department of the AIR FORCE, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a ballastable, towed, pneumatic tired, oscillating wheel hopper body roller.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Military

MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Material.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-H-15424	- Hand tools; Packaging of.
MIL-W-52574	- Welding and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels.
MIL-T-52583	- Tire, Pneumatic, Large Size, Off-the-Road, General Specification for.

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STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MS51337	-	Coupler, Drawbar, Ring - Heavy Duty.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 Description. The roller shall consist of a steel hopper body, equipped with 13 pneumatic tired, oscillating-wheels, with the front set mounted on a bolster. The roller shall be capable of being towed at speeds up to a maximum of 15 miles per hour (mph), carrying the maximum gross load. The overall dimensions and weights of the roller shall be as follows:

Length, with tongue and lunette, (maximum)	195 inches
Width, (maximum)	88 inches
Height, (maximum)	48 inches
Weight, (empty)	3,300 lbs.
Weight, (maximum ballastable)	18,000 lbs.
Cubic volume of hopper	115 cu. ft.

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3.2 Preproduction model. The supplier shall furnish a roller for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce rollers that will comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government a roller for inspection as specified in 4.5.

3.4 Material. Material shall be as specified herein. All material shall be new and unused. Material not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering standards of the earth moving and construction equipment industry and as set forth in the publications of the American Society for Testing and Materials, Society of Automotive Engineers including SAE Handbook, and the United States of America Standards Institute. Rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the roller by the Government. Threaded parts shall be made in accordance with SAE Standards. Maximum practicable use shall be made of interchangeable hardware and fastening devices and a minimum number of types and sizes of screws, bolts, nuts, washers and similar common (standard) parts.

3.5 Maintainability. All roller components shall be accessible for maintenance and repair utilizing common hand tools. The replacement and adjustment of all components shall be accomplished with minimum disturbance to other components. The tires and wheels shall be mounted to permit accessibility to tire inflation valves and wheel oscillating pivot shafts.

3.6 Body. The body shall be fabricated of not less than 3/16 inch thick steel plate which shall be structurally reinforced with steel members to form a rigid ballast hopper box. Seams and reinforcing structural members of the ballast hopper box shall be continuously welded so as to form a watertight box. The ballast capacity of the hopper box shall conform to 3.1. The hopper box shall be capable of carrying a load of gravel, sand, earth, water, or any combination thereof, which will be equivalent to the difference between the empty weight and the gross weight as specified in 3.1. A clean-out door having an area of not less than 100 square inches shall be located on each side of the hopper box. Bolts shall be tack welded to the hopper box from the inside. The clean-out doors shall be positioned near the corners of the hopper box and shall be placed diagonally. No obstruction shall block the drainage of water to the clean-out doors.

3.7 Bolster. The front bolster shall be of the yoke type with three legs of cast steel or heavy fabricated steel plate. The bolster shall be attached to the body by a fifth wheel and kingpin permitting a lateral swing of 260 degrees. The bolster shall also rotate about an axis parallel with the center line of travel.

3-8. Tongue. The tongue shall be fabricated from steel and shall resist the strains of pulling, backing, and turning without stress fatigue or deformation when tested as specified in 4.7.2.3. The tongue shall be attached to the bolster by removable pins and anchor sockets. The pins will be fastened to the bolster by chains not exceeding 10 inches in length. The chains shall be welded to the bolster in such a position that ample working space is provided when inserting pins in tongue and anchor socket. When installed, the pins shall be secured by spring loaded wire clamps. The tongue shall be free to move from ground level through an arc not to exceed 80 degrees above the horizontal. The tongue shall have a lunette conforming to MS51337.

3-9. Wheels, wheel hubs, tires, and tubes.

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3-9.1 Wheels. The 13 wheels shall be mounted on axles which shall be located on the same lateral axis with 6 wheels in front and 7 wheels in the rear. Wheels shall be of the demountable type and shall revolve and oscillate in parts independent of the adjacent pairs on an axis parallel with the line of travel, except for one rear wheel, which shall be either rigidly mounted or independently mounted. The seven rear-mounted wheels shall overlap the tracking of the six front mounted wheels. Minimum overlap for each wheel shall be 1/4 inch. Maximum overlap shall be 1/2 inch. The axle pivot housings shall be fastened to the support brackets with steel pivot shafts.

3-9.2 Wheel hubs. Wheel hubs shall be fitted with roller bearings suitable for the required speed and loads. A positive method of making minor adjustments to the bearings shall be provided.

3-9.3 Tires. Tires shall have a smooth tread, for compactor use and shall conform to MIL-T-52583, Code C-1, style TT. Tire size shall be not less than 7.50 by 15, 6 ply rating. Except for compound systems inherently weather-resistant, commercial or military antiozonant compounds shall be added to retard ozone-induced tread and sidewall cracking.

3-9.4 Tubes. Tubes shall conform to MIL-T-52583.

3-10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubrication.

3-10.1 Lubricants. The roller shall operate as specified herein when lubricated with grease conforming to MIL-G-10924.

3-10.2 Lubrication fittings. Lubrication fitting shall conform to MIL-F-35412, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a flexible hose and hydraulic coupler. Accessibility to fittings shall be provided with out the removal or adjustment of other components.

3-10.3 Pressure release device. An automatic pressure release shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3-10.4 Grease lubrication. Grease lubrication shall be with grease conforming to MIL-G-10924. The roller shall be assembled, run in, tested, and delivered with this grease. Bearings shall be cleaned before lubrication because military grease is not compatible with other greases. The roller shall be tagged in a conspicuous place to indicate that military grease has been used.

3-11 Lifting attachments. Permanently affixed attachments that enable the empty roller to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the roller, 2-1/2 times the resultant vectorial force imposed when the roller is suspended by the lifting attachments. Attachments for multiple slings shall be located so that (a) slings will converge not more than 19 feet above the lowest extremity of the roller, and (b) not less than 1-inch clearance will be maintained between the slings and the roller. The inside diameter of the attachment eye shall be not less than 3 inches. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.12.

3-12 Tiedown attachments. Permanently affixed attachments that permit fastening the empty roller to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the roller, its proportion of the following loads applied to the roller: 8-G forward, 8-g rearward, 2-1/4-G upward, 1-1/2-G sideward, where G equals cargo weight. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.11.

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3-13 Stenciling. The empty weight of the roller and the maximum tire pressure shall be stenciled on each side so as to be discernible. The words "LIFT" and "TIEDOWN", as applicable, shall be stenciled adjacent to their respective attachments using block letters not less than 1-inch high. Stenciling shall conform to MIL-T-704.

3-14 Treatment and painting. The portions of the roller normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3-15 Identification marking. The roller shall be identified in accordance with MIL-STD-130. The identification plate shall conform to MIL-P-514, type I, style 1, composition C, grade A, class 1.

3-16 Instruction plate. An instruction plate conforming to MIL-P-514, type III, composition C, with grade A, class 1 material, shall be furnished. The plate shall include the shipping data and a silhouette of the roller indicating the center of gravity and the location and capacity of each lifting and tiedown attachment. The plate shall be secured to the roller in a conspicuous location by bolts, rivets, or screws.

3-17 Technical publication. Such technical publications as are specified shall be furnished (see 6.2).

3-18 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3-19. Toolbox. A toolbox shall be provided and shall be of a size to hold those maintenance tools specified in 3.18. The toolbox shall be made of steel 0.0747-inch nominal thickness and shall include provisions for drainage in its mounted position. The toolbox shall have a hinged lid with the hinge extending the full length of one edge and shall be provided with not less than two spring tension fasteners that will keep the lid closed when the toolbox is subjected to vibration. The toolbox lid shall have an overlapping lip to prevent entry of rain and dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices.

3-20 Workmanship. All components and assemblies of the roller, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All exposed outer edges shall be rounded or chamfered.

3-20.1 Welders qualification. Welder qualification shall be in accordance with MIL-W-52574.

3-20.2 Welding. Welding of steel shall be in accordance with MIL-W-52574, welding process optional, except that joint welding procedures are not required for type 1 steels.

4. QUALTITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Component and material inspection. The supplier is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.3 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.4).
(b) Initial production inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).
(d) Inspection comparison (see 4.8).
(e) Inspection of preparation for delivery (see 4.9).

4.4 Preproduction inspection.

4.4.1 Examination. The roller shall be examined as specified in 4.7.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The roller shall be tested as specified in 4.7.2.3 through 4.7.2.7. Failure of any test shall be cause for rejection.

4.5 Initial production inspection.

4.5.1 Examination and tests. When specified (see 3.3) an initial production roller will be selected at random by the Government from the first five rollers being produced from production tooling and will be examined as specified in 4.7.1 and tested as specified in 4.7.2.3 through 4.7.2.7 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production roller shall not exclude the remaining rollers from the quality conformance inspection and acceptance provisions specified in section 4.

4.5.2 Inspection failure. Failure of an initial production roller to meet any requirement specified herein during and as a result of the examination and test specified in 4.5.1 shall be cause for rejection of the initial production roller (s) and shall be cause for refusal by the Government to continue acceptance of production rollers until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on rollers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all rollers accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.6 Quality conformance inspection.

4.6.1 Examination. Each roller shall be examined as specified in 4.7.1. Presence of one or more defects shall be cause for rejection.

4.6.2 Test. Each roller shall be examined as specified in 4.7.2.2. Failure of the test shall be cause for rejection.

4.7 Inspection procedure.

4.7.1 Examination. The roller shall be examined for the following defects:

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101. Dimensions or weight not as specified.
102. Maintainability not as specified.
103. Material not as specified.
104. Body not as specified.
105. Clean-out doors not as specified.
106. Tongue not as specified.
107. Axles, wheels, rims, tires, or tubes not as specified.
108. Lubrication fittings not as specified or not accessible.
109. Stenciling not as specified.
110. Treatment or painting not as specified.
111. Identification marking not as specified.
112. Shipping data instruction plate not as specified.
113. Technical publications missing or not as specified.
114. Repair parts missing or not as specified.
115. Maintenance tools missing or not as specified.
116. Toolbox not as specified.
117. Workmanship not as specified.
118. Welding, including material identification and procedures, not as specified.

4.7.2 Tests.

4.7.2.1 Test conditions. Prior to testing, the roller shall be lubricated with grease as specified in 3.10 and all tires shall be inflated to 35 psi using a standard tire inflation chuck.

4.7.2.2 Performance. Tow the roller, unballasted, for one hour. Start and stop the roller four times. Swing the roller to the right and to the left within the range specified in 3.7. Back the roller to the right and to the left within the range specified in 3.7. Inability of the roller to complete the test, binding of the bolster, inability to perform turns of the range specified in 3.7, or damage to the fifth wheel, kingpin, or any structural component shall constitute failure of the test.

4.7.2.3 Operational. The roller shall be subject to a 20 mile operational test. This test shall be conducted with the roller ballasted with wet sand to equal the maximum ballastable weight as specified in 3.1. When practicable, the speed shall be maintained at 15 miles per hour (mph) during the test. Examine the tongue in various positions for structural damage or permanent deformation. Examine the tongue connecting pins for compatible size, ease of placing, removal, and the capability of being locked in position. Observe the overlap of the tracks of the seven rear wheels with the tracks of the six front wheels. Immediately after completion of the road test, examine the wheel bearings for over heating and indications of damage to the bearings, bearing cages, races, or grease retainers. Inability of the roller to accept the lubrication specified in 3.10, failure to complete the 20 mile road test, failure to comply with the maximum ballastable weight, any indication of stress fatigue or deformation to the body, tongue, structural assembly, nonconformance to 3.8 or 3.9.1, or wear or over heating or damage of any wheel bearing, or tires, wheels or hubs shall constitute failure of the test.

4.7.2.4 Compaction. Operate the roller at the maximum ballastable weight as specified in 3.1 over uncompacted soil for 8 hours. Then operate the roller both transversely and longitudinally on a grade of 4 to 6 percent for one hour. Turn the roller to the maximum angle right and left as specified in 3.7 while moving forward and reverse. Failure of the roller to complete the 8 hour compaction test, any sign of inability to turn to the maximum angle right and left while moving forward and reverse, or any signs of excessive wear, structural deformation, or damage to any component of the roller shall constitute failure of this test.

4.7.2.5 Watertightness. Fill the roller with water and examine for watertightness. Drain the roller through the clean-out doors and clean the roller body. Inability to perform this test shall constitute failure of the test.

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4.7.2.6 Lifting attachments. Suspend the empty roller by the lifting attachments with multiple slings. Determine the load on each lifting attachment with the roller in suspension. Lower the roller and subject each lifting attachment to a pull equal to or greater than 2-1/2 times the vectorial load reading previously obtained on these lifting attachments while the roller was suspended. Nonconformance to 3.11 shall constitute failure of this test.

4.7.2.7 Tiedown attachment. Determine the empty weight of the roller. Subject each tiedown attachment to its proportionate share of the forward, rearward, upward and sideward loads specified in 3.12. Measurements shall be made with a dynamometer scale or other accurate measuring device. Nonconformance to 3.12 shall constitute failure of this test.

4.8 Inspection comparison. The Government may select rollers at any time during the contract production period and subject these rollers to the examination specified in 4.7.1 and the tests specified in 4.7.2.3 through 4.7.2.7 to determine that the quality of the selected rollers is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Rollers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production roller. Failure of any roller when so inspected may be considered cause for refusal by the Government to continue acceptance of rollers until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which cause the rejection.

4.9 Inspection of preparation for delivery.

4.9.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.9.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.9.3 Examination. Samples selected in accordance with 4.9.2 shall be examined for the following defects. AQL shall be 2.5 percent defective, inspection level, S-2.

119.	Materials, methods and containers not as specified. Each incorrect material, method or container shall be considered one defect.
120.	Unprotected exterior metal surfaces not coated with preservative as specified.
121.	Tires not inflated as specified.
122.	Maintenance tools not preserved in accordance with the referenced document.
123.	Intermediate packaging not as specified.
124.	Marking missing, illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The tongue shall be detached from the bolster.

5.1.1.2 Unprotected surfaces. Unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservation shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Tires. Tires shall be inflated to 10 pounds above the specified operating pressure.

5.1.1.4 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

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5.1.1.5 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of MIL-H-15424.

5.1.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.7 Intermediate packaging. The repair parts, maintenance tools, technical publications and drain plugs shall be intermediate packaged in the toolbox, if space permits, and the toolbox lid secured to prevent pilferage. When there is not sufficient space, the items shall be intermediate packaged in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure shall be in accordance with the appendix to the box specification, except strapping shall not be required.

5.1.2 Level C. Each complete roller shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard commercial practice is acceptable when it fulfills these requirements.

5.2 Packing. The roller, with the tongue and intermediate package placed and secured in the body, shall be prepared for shipment mobile in a manner to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules (see 6.4).

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rollers are intended for compaction of earth fill, sand, gravel or asphaltic concrete pavements.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of production model (see 3.2).
(c) When initial production roller is required (see 3.3).
(d) Technical publications required (see 3.17).
(e) Repair parts and maintenance tools required (see 3.18).
(f) Level of preservation and packaging required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production rollers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish rollers conforming to this specification.

6.4 Levels of packing have not been included, as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing level marking, the level of packing shall be considered level A.

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